AfyaPlus AI Triage — Cost-Controlled Pilot Decision

Recommendation: GO for a limited technical pilot; NO-GO for clinical use.

Unit economics

Measure	Baseline	10× traffic scenario
Requests/month	10,000	100,000
Model cost / 1,000	$0.105	$0.105
Total / 1,000 incl. fixed allocations	$5.905	$0.685

The 10× total is an illustrative allocation of unchanged $18/month infrastructure and $40/month operations across more requests—not a claim that capacity costs remain fixed under real traffic.

Two optimisation levers

1. Exact-response cache: 30 fixed fixture requests (3 repeated cases × 10 rounds) produced 27 hits: 90% hit rate and 27 model calls avoided. This repetitive fixture is not representative traffic.

2. Compact response path: output ceiling falls from 180 to 80 tokens (55.6% lower ceiling). This is a token-cap proxy, not quantization; measure actual tokens and review quality.

10× traffic plan

Bound input/output; cache exact repeats; monitor latency, errors and spend; use provider rate limits; load-test before adding replicas; document a degraded-mode response. Budget alerts are not hard spending caps.

Main risk and mitigation

Shortened or cached health guidance can be incomplete or stale. Keep this as a fictional-data educational demo; require clinician-led evaluation, privacy/security review and an approved fallback before any clinical use.

Decision conditions

Proceed only with a $5 monthly budget alert, usage monitoring, bounded tokens, secrets kept out of source control, and an owner for alerts. Revisit self-hosting only when measured sustained volume and credible capacity quotes support the break-even analysis.

Assumptions: 450 input + 120 output tokens/request; $0.10/M input and $0.50/M output (official price snapshot checked 2026-10-14; re-verify); $18/month infrastructure; 2 ops hours/month × $20/hour; 10,000 baseline requests/month. Infrastructure and staffing are estimates, not quotes. Mock fixture results are not production benchmarks.